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1986 Interpretations—Technical Publications

Answers to Technical Inquiries

DECEMBER 1987

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TITLE 8.1 0732290 0058069 1

1986 Interpretations—Technical Publications

Answers to Technical Inquiries

Industry Affairs

DECEMBER 1987

American Petroleum Institute



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Issue Date: September 15, 1986
Publication: Standard 610, Centrifugal Pumps for General Refinery Services,
Edition/Date: Sixth Edition, 1981
Reference: 610-I-02/86

Question: Please explain the reasoning behind what appear to be very low allowable forces and moments. Point out in detail those "weak" points to pay special attention to when considering loads more than twice those shown in Table 2.

Reply: The values in Table 2 are the result of many years of experience and testing, and represent the highest values the manufacturers can consistently provide at reasonable cost. Pumps that can withstand higher forces and moments are available, and can be provided when required but at additional cost. The controlling part of the system is usually the baseplate, which must resist the forces and moments of Table 2 sufficiently to limit the deflection at the coupling to 0.005 inch. Loads greater than these are permissible if the coupling can accomodate the greater deflection, and are suggested in Appendix C. Note that paragraph 2.4.2 specifies that the pump case must withstand twice the forces and moments in Table 2.

The manufacturers can comply with paragraph 2.4.3 at reasonable cost, and those pumps will be satisfactory for most applications. If piping calculations indicate that a greater stiffness in the pump support is required, a baseplate capable of withstanding twice the loads is available from any manufacturer at extra cost (see paragraph 2.4.6). By working closely with the manufacturers, pumps able to withstand even greater loads are usually available. The intent is to spend only as much as necessary for each application; but to have options available to meet higher imposed loads when that is the best economic choice.

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Issue Date: January 15, 1986

Publication: Manual of Petroleum Measurement Standards, Chapter 12.2, "Calculation of Liquid Petroleum Quantities Measured by Turbine or Displacement Meters"

Edition/Date: First Edition, September 1981

Question: Is the procedure in 12.2.7.4 of Chapter 12.2 correct?

Reply: This response confirms that the procedure in 12.2.7.4 is correct. What follows is an interpretation of that procedure compiled from responses provided by API technical experts.

The basic meter factor equation is as follows, for a non-temperature compensated meter:

$$\text{Meter factor} = \frac{\text{Base prover volume} \times CTSP \times CPSP \times CTLP \times CPLP}{\text{Indicated meter volume on proving run} \times CTLM \times CPLM}$$

Since the open prover tank is at 0 psig, the CPSP and CPLP would both be 1.0000. The meter, being under pressure, would have a CPLM of some value greater than 1.0000. Therefore:

$$\text{Meter factor} = \frac{BVP \times CTSP \times 1.0000 \times CTLP \times 1.000}{MV \times CTLM \times CPLM} = \frac{BPV \times CTSP \times CTLP}{MV \times CTLM \times CPLM}$$

Even so, the measurement ticket has to be considered:

MF x CTL (for delivery temperature on meter)
x CPL (for delivery pressure at meter)
x Indicated volume for the delivery

= MF x CTLD x CPLD x IVD

= Standard volume delivered.

To illustrate:

$$(\text{Meter factor}) \times CTLD \times CPLD \times IVD$$

$$= \frac{(BPV \times CTSP \times CTLP)}{(MV \times CTLM \times CPLM)} \times CTLD \times CPLD \times IVD$$

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1986 INTERPRETATIONS

Issue Date: June 9, 1986
Publication: Specification Q1, Specification for Quality Programs
Edition/Date: First Edition, January 1985

Question 1: Is it mandatory that all job positions listed in the organization section of a manufacturer's Quality Assurance Manual (i.e., president, directors, managers, supervisors) meet the full requirements of Specification Q1, paragraph 3.3.1?

Reply: Paragraph 3.3.1 has been revised to require documented personnel qualifications only for personnel performing activities addressed in the quality program (3.4 through 3.20). See Specification Q1, Supplement 1, issued June 1, 1986. In addition, the Task Group on Specification for Quality has recommended a change to 3.2.1 to require a document organization only for each position with functions in the quality program.

Question 2: Is it mandatory that all measuring and test equipment used in the manufacturing of products that exhibit the API monogram (i.e. those used by machinist or maintenance personnel) meet the full requirements of Specification Q1 paragraph 3.1.4.1?

Reply: The Task Group on Specification for Quality is also recommending a change to 3.13.1 to restrict the inspection/testing for acceptance. The measuring and testing equipment of 3.14.1 would only be that dealing with product acceptance. The use of uncalibrated measuring and testing equipment other than that for product acceptance is an economical decision and outside the scope of API.

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Issue Date: July 21, 1986
Publication: Standard 610, Centrifugal Pumps for General Refinery Services
Edition/Date: Sixth Edition, 1981
Reference: 610-I-03/86

Question: Reference paragraph 2.1.7. Define stable head-capacity curve. Why is it preferable to have 10-20% head rise to shut off for parallel operation? Discuss effects of pumps with less than 10% head rise, in parallel and in single operation, with and without control valves.

Reply: A stable head-capacity curve has only one defined flow rate for each value of head. Maximum 20% head rise is to limit the maximum pressure imposed on the system. Minimum recommended head rise of 10% is to minimize flow imbalance between pumps operating in parallel, and especially to ensure stable head-capacity curves for pumps in parallel or if a back pressure control valve is in the system.

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Issue Date: December 30, 1985
Publication: Standard 614, Lubrication, Shaft-Sealing, and Control Oil Systems for Special-Purpose Applications
Edition/Date: Second Edition, 1984
Reference: 614-I-01/85

Question 1: The wording of 4.4.2.2 and 4.4.2.3 in the second edition is as follows:

4.4.2.2 All exterior machined surfaces shall be coated with a suitable rust preventive that is compatible with the normal charge oil or that can be completely removed by flushing with the normal charge oil.
4.4.2.3 All internal steel surfaces of components fabricated of carbon steel, such as vessels, pumps, and piping, shall be coated with a suitable oil-soluble rust preventive.

We believe this is incorrect and that the wording should be altered as follows:

4.4.2.2 All exterior machined surfaces shall be coated with a suitable oil-soluble rust preventive.
4.4.2.3 All internal steel surfaces of components fabricated of carbon steel, such as vessels, pumps, and piping, shall be coated with a suitable rust preventive that is compatible with the normal charge oil or that can be completely removed by flushing with the normal charge oil.

Reply: The wording of paragraphs 4.4.2.2 and 4.4.2.3 in the second edition is not correct. The wording will be changed similar to your suggestion in the next revision.

Question 2: Please clarify the terms "console," "packing," and "multiple-package."

Reply: The terms "console" and "package" are almost identical in meaning. We think of a package as a single unit; in the case of an oil system, it is one baseplate on which is mounted all of the components. None of the components are shipped to the plant site separately. Therefore, the purchaser does not have to do any on-site assembly of components. In many cases, however, a single skid or baseplate is not possible. The manufacturer, may with the agreement of the purchaser, supply the system in "multiple-package arrangement". For example, it may consist of two baseplates which bolt together at the plant site. The

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Issue Date: March 7, 1986
Publication: Standard 598, Valve Inspection and Test
Edition/Date: Fifth Edition, 1982
Reference: 598-I-01/85

Question 1: Does API Standard 598 cover the resilient seated gate valve type commonly referred to as "double block and bleed"?

Reply: Yes, see 1.1.1.

Question 2: Must the inlet seat show no leakage?

Reply: Testing is conducted by one of three options listed in 4.4.2 as for a regular gate valve. Gate valves are considered to be "designed to close against pressure from both directions" unless the manufacturer so states and marks on the valve body a undirectional sealing feature.

Question 3: Where gate valves have been fitted with resilient seating in the seat rings or wedge only, and are not of the "double block and bleed" style, does API Standard 598 require that the seat closure tests show no leakage at the upstream side of the valve?

Reply: All three options in 4.4.2 require both seats of gate valves to be tested. The only exception would be if the valve had the undirectional feature described in 2 above.

Question 4: Does the maximum allowable leakage rate described refer to the downstream side of the valve where the wedge would naturally seal, or does it also apply to the upstream side?

Reply: The 0 leakage rate for "all resilient seated valves" specified in Table 5 applies to all test options in 4.4.2.

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Issue Date: September 5, 1986
Publication: Specification Q1, Specification for Quality Programs
Edition/Date: First Edition, January 1985

Question: Paragraph 3.18.1 addresses scheduled and unscheduled internal audits. Does each audit, scheduled and unscheduled, have to audit all criteria of the quality program? Does the combined scheduled and unscheduled internal audit that verifies compliance annually fulfill the requirements of 3.18.1?

Reply: The combined scheduled and unscheduled internal audits that verify compliance annually of all criteria of the quality program fulfill the requirements of 3.18.1.

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Issue Date: October 27, 1986
Publication: Standard 598, Valve Inspection and Test
Edition/Date: Fifth Edition, 1982
Reference: 598-I-01/86

Question: When air is used as a test fluid for the shell test of Standard 600 valves required by Standard 598, Fifth Edition, paragraph 3.2.1, is it permissable to test at 60 to 100 psi (5 to 7 bar)?

Reply: No. Per ANSI B16.34-1981 (referenced in Table 2 of API 598), paragraph 7.1, the test pressure shall be "no less than 1.5 times the 100 F rating, rounded off to the next higher 25 psi increment."

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Issue Date: September 5, 1986
Publication: Specification Q1, Specification for Quality Programs
Edition/Date: First Edition, January 1985

Question: Paragraph 3.3.1 on qualification of personnel is very clear when one is to identify individuals that must have documented evidence of qualification while performing design verification (3.6.3), inspection/testing (3.13.2), and audits (3.18.1). Do personnel performing review and approval of instructions, procedures, specifications, and drawings (3.7.2) require documented evidence of qualifications? Are personnel responsible for approval of process control (3.8.3) also required to have documented evidence of qualifications? Are personnel reviewing and approving procedures for special processes (3.12.2) required to have supporting qualification documentation? Are there any other groups of individuals, other than those addressed above, that would require documented evidence of qualifications?

Reply: Paragraph 3.3.1 requires that all personnel authorized to conduct activities addressed in 3.4 through 3.20 be qualified to perform their assigned activity. Where API specifications dictate specific qualification requirements and methods, they should be followed. For activities where API specifications do not state specific personnel qualifications or methods, the manufacturer needs to establish, document, and follow his own.

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Issue Date: July 21, 1986
Publication: Standard 600, Steel Gate Valves, Flanged and Buttwelding Ends
Edition/Date: Eighth Edition, 1981
Reference: 600-I-03/85

Question: With reference to paragraph 2.7.1 and Table 2, is the minimum stuffing box depth for a Class 150 valve with a nominal 7/8 inch stem diameter 6 X nominal packing width (i.e., 1-1/2 inches)? 6 X actual packing ring width new? or 6 X actual packing width compressed?

Reply: The minimum stuffing box depth for this example is 6 X nominal packing width (1/4 inch) or 1-1/2 inches.

Question 2: With reference to paragraph 2.7.1 and Table 2, if six rings of packing can be fitted in a box less than 1-1/2 inches in depth, is this satisfactory?

Reply: No. It would not meet the requirements of 2.7.1 and Table 2. However, API 600 is currently under review for a new edition. The subject of stuffing box dimensions is on the agenda for the new edition.

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Issue Date: July 21, 1986
Publication: Standard 600, Steel Gate Valves, Flanged and Buttwelding Ends
Edition/Date: Eighth Edition, 1981
Reference: 600-I-04/85

Question 1: Does the reference to B16.5 in paragraph 2.2.3 on spotfacing or backfacing refer only to parallelism of the flange faces or does it also mean that B16.5 flange thickness requirements apply to bonnet and body to bonnet flanges?

Reply: The reference to B16.5 in paragraph 2.2.3 on backfacing or spotfacing refers only to parallelism of the flange faces.

Question 2: If B16.5 flange thicknesses do not apply to bonnet and body to bonnet flange thicknesses, what are the minimum bonnet and body to bonnet flange thicknesses?

Reply: API 600 does not address the thicknesses for bonnet and body to bonnet flanges.

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1986 INTERPRETATIONS

Issue Date: June 25, 1986
Publication: Manual of Petroleum Measurement Standards, Chapter 8.1, "Manual Sampling of Petroleum and Petroleum Products"
Edition/Date: First Edition, October 1981

Question: Please clarify the phrase "when required," that is, should ship's tank samples be tested for sediment and water if the automatic pipeline sampler is operating properly?

Reply: From a technical point of view, it seems reasonable to interpret the paragraph as meaning "when required" for technical or operation reasons. This could mean when the automatic pipeline samplers are not operating properly or when the sampler does not meet recommended performance standards as recognized by two parties or a shore tank sample is unavailable.

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Issue Date: August 22, 1986
Publication: Standard 600, Steel Gate Valves, Flanged and Buttwelding Ends
Edition/Date: Eighth Edition, 1981
Reference: 600-I-05/85

Question 1: With respect to paragraph 2.2.4, is it permissible to calculate the tensile stress area of the bolts, rather than the total root cross sectional area of the bolts, when calculating the bonnet joint bolting stresses?

Reply: No. Paragraph 2.2.4 requires that the total root cross section area be used in calculating bonnet joint stresses.

Question 2: With respect to paragraph 2.2.4, what are the reasons for specifying through bolts for the bonnet joints and what are the objections, if any, to the use of studs for bonnet joints?

Reply: Please be advised that API provides responses to requests for clarification and interpretations of API standards and considers suggestions for revisions. API normally does not provide rationale for requirements since the requirements are based on consideration of technical data and experience and expertise of the individual committee members.

Question 3: With respect to paragraph 3.3.1 and 3.11, why does the API not include graphite as an acceptable gasket material and stem packing material?

Reply: The API is currently considering a proposed change to its valve standards which will address alternatives for acceptable gasket and packing material.

Question 4: With reference to paragraph 2.7.1 and Table 2, would we be complying with API if we changed the stuffing box dimensions to suit graphite packing?

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interconnecting piping between the two baseplates may be designed to bolt together or it may require the purchaser to provide interconnecting piping.

A complete single or multiple package arrangement constitutes a console. Obviously the oil system might contain components such as overhead seal oil tanks which have to be shipped separately. Such a component would be considered as one package in a multiple-package arrangement.

Question 3: Describe pipe bushings, unions, and couplings, and the intent of 2.13.15.

Reply: The intent of paragraph 2.13.15 is described in 2.13.14, which is "to minimize the use of flanges and fittings". The object is to keep to a minimum potential sources of oil leaks and places where dirt can be trapped.

Question 4: Explain what a "bronze hose with four-wall interlocking construction referenced in 3.5.7 is.

Reply: The confusion is over the term "four-wall interlocking construction". The hose is simply a flexible conduit made of spiral wound bronze strip.

Question 5: What is the difference between "bill of material" (5.2.2.4.g) and "parts list" (5.2.3.3)?

Reply: The bill of material lists all components being supplied; for example, a pump. The pump manufacturer would be required to supply a parts list for that pump showing its components such as bearings, seals, etc.

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Reply: No. Paragraph 2.7.1 and Table 2 require that the stuffing box dimensions be as specified. However, API 600 is currently under review for a new edition. The subject of stuffing box dimensions for graphite packing will be considered.

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Issue Date: March 7, 1986
Publication: Standard 600, Steel Gate Valves, Flanged and Buttwelding Ends,
Edition/Date: Eighth Edition, 1981
Reference: 600-I-01/86

Question: Can valves designed with minimum wall thicknesses in accordance with ANSI B16.32, paragraph 6.1.1 and Table 3 be certified to meet the requirements of API Standard 600, Eighth Edition, January 1981?

Reply: No. Wall thicknesses of Standard 600 valves must comply with Standard 600, paragraph 2.2.1 and Table 1. Based on experience in the refining industry, API has had this requirement for many years.

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Issue Date: August 6, 1986
Publication: Standard 600, Steel Gate Valves, Flanged and Buttwelding Ends,
Edition/Date: Eighth Edition, 1981
Reference: 600-I-03/86

Question: Does ASTM A487, Grade CA6NM meet the requirement for Trim Nos. 1, 4, 6, 7, 8 and 8a, Table 3 of API Standard 600?

Reply: The materials listed in Table 3 of API 600 are typical specifications and represent some acceptable grades. ASTM A487, Grade CA6NM is generally accepted as equal to ASTM A215 Grade CA15 in corrosion resistance. However, a purchaser may have valid reasons for specifying a certain grade of nominal trim material. As written in Notes to Purchaser in item 2n on page vii, nominal trim material is identified as an item that is to be specified by the purchaser.

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Question 6: Does the term "valves" in 5.2.3.1 comprise all sorts of valves or only "control valves" and "relief valves"?

Reply: Reference to "valves" means control and relief valves.

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Issue Date: January 21, 1986
Publication: Standard 607, Fire Test for Soft-Seated Ball Valves
Edition/Date: Third Edition, 1985
Reference: 607-I-03/85

Question 1: What is the test pressure for testing Class 800 screwed and socket weld ball valves per API Standard 607, Second Edition, August 1980?

Reply: The Second Edition of API Standard 607 did not specifically cover Class 800. The Third Edition of API Standard 607, dated November 1985 has a revised procedure. The test pressure during the fire test is 75% of the cold working pressure and 7% of the cold working pressure after the cool down period.

Question 2: What other pressure classes are qualified with the successful test of a Class 800 valve?

Reply: Per the Third Edition, a successful test of a Class 800 valve qualifies valves of the same design for Classes 900 and 1500.

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Issue Date: August 22, 1986
Publication: Standard 620, Recommended Rules for Design and Construction of Large, Welded, Low-Pressure Storage Tanks
Edition/Date: Seventh Edition, 1982
Reference: 620-I-03/85

Question: why were the requirements for minimum charpy V-notch impact valves increaed in Table R-3 of API Standard 620, and what is the technical background and analytical data which brought these new requirements?

Reply: The higher toughness values more realistically represent what is achievable from the materials specified.

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Issue Date: March 7, 1986
Publication: Standard 607, Fire Test for Soft-Seated Ball Valves
Edition/Date: Second Edition, 1980
Reference: 607-I-05/85

Question: Is there a specific size range this (paragraph 1.3.5) covers or is it merely a statement which covers any valve bodies which reach the 1,100 F minimum in less than 20 minutes while complying with the other requirements of the fire exposure?

Reply: There is no specific size range covered by paragraph 1.3.5. The statement merely recognizes that small valves heat quicker than large valves. Because of this, the valve body may exceed the minimum 1,100 F by a wide margin and even approach flame temperature. Therefore, this section allows the option of reducing flame temperature, so long as the valve body is maintained above 1,100 F; and the time at this temperature is a minimum of 10 minutes. If these two criteria are met, then the requirements of the fire exposure are met, even though the overall time frame is less than the normal 30 minutes stipulated in 1.3.3 and 1.3.4.

The Third Editon of Standard 607 was published in November 1985 and this paragraph has been deleted from the standard.

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E-07-21

Issue Date: August 6, 1986

Publication: Standard 620, Recommended Rules for Design and Construction of Large, Welded, Low-Pressure Storage Tanks

Edition/Date: Seventh Edition, 1982

Reference: 620-I-04/85

Question: May materials without impact requirements be used for an internally insulated suspended deck in an API Standard 620 Appendix R tank?

Reply: Appendix R does not provide material requirements for internally insulated suspended decks.

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Issue Date: August 6, 1986
Publication: Standard 620, Recommended Rules for Design and Construction of Large, Welded, Low-Pressure Storage Tanks
Edition/Date: Seventh Edition, 1982
Reference: 620-I-01/86

Question: In Figure 3-4 (f) of API Standard 620, the weld between the thickened shell plate and the thickened roof plate is an apparent full penetration weld. Is a full penetration weld a requirement?

Reply: No.

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Issue Date: September 8, 1986
Publication: Standard 607, Test for Soft-Seated Ball Valves
Edition/Date: Third Edition, 1985
Reference: 607-I-01/86

Question: Is Standard 607 applicable to a 3-way ball valve? Will Standard 607 qualify 3-way ball valves?

Reply: Yes, Standard 607 applies to 3-way ball valves and will qualify such a valve in the fire test. It should be noted that additional plumbing will be needed so that leakage from all seats can be measured. This will probably require repetitive testing in each port shutoff configuration.

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Issue Date: April 17, 1984
Publication: Standard 610, Centrifugal Pumps for General Refinery Services
Edition/Date: Sixth Edition, 1981
Reference: 610-I-4A-84

Question: Please explain the purposes of test tolerances on horsepower and efficiency.

Reply: The test tolerances on horsepower and efficiency are for two different purposes. While it may seem that meeting one tolerance would ensure meeting the other tolerance, that is not necessarily true. The reasons for each tolerance are:

1. The tested horsepower (at rated flowrate and at the rated head plus or minus the tabulated tolerances) may not exceed the rated horsepower by more than 4 percent. Since the horsepower rating of the driver selected might only exceed the pump rated horsepower by 10 percent (see 3.1.2), the 4 percent tolerance on horsepower assures 6 percent reserve driver horsepower to handle occasionally higher pump flowrates and/or degradation of pump efficiency. This positive tolerance on horsepower is necessary to permit the positive tolerance on head without necessarily requiring higher pump efficiency to compensate.

2. The tolerance on tested efficiency is -1/2 percent to allow for rounding to the nearest whole percent. It is intended that the rated efficiency be a guaranteed value and any deviation be positive, except for the rounding allowance.

The tolerances on these two characteristics cannot be reduced to one characteristic because the two are not necessarily met coincidentally. For example, with low head pumps where the head tolerance is -2 percent to +5 percent, the efficiency will have to be greater than rated in order to meet the +4 percent horsepower tolerance if the head is 5 percent higher than rated.

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E-07-21

Issue Date: August 6, 1986
Publication: Standard 620, Recommended Rules for Design and Construction of Large, Welded, Low-Pressure Storage Tanks
Edition/Date: Seventh Edition, 1982
Reference: 620-I-02/86

Question 1: Does API Standard 620 Appendix R require the meridional or latitudinal joints between shell plates, 40-45 mm in thickness but not having nozzles or other welded attachment, to be postweld heat treated?

Reply: No.

Question 2: Does API Standard 620 Appendix R require the meridional or latitudinal joints in the compressional ring region having plates 50 mm in thickness, similiar to Figure 3.4 (g), to be postweld heat treated?

Reply: No.

Question 3: Does the maximum underrun in paragraph 2.2.4.1 mean a maximum of 0.01 inch from the required thickness (3.10.3), and a maximum of 0.01 inch from the least permissible thickness (3.10.4)?

Reply: Yes.

Question 4: Can the maximum permissible undercutting of paragraph 4.13 be measured from the surface of plates having a thickness underrun of 0.01 inch?

Reply: Yes.

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Issue Date: December 30, 1986
Publication: Standard 620, Recommended Rules for Design and Construction of Large, Welded, Low-Pressure Storage Tanks
Edition/Date: Seventh Edition, 1982
Reference: 620-I-03/86

Question: Can subsonic vibrations be substituted when stress relieving of plates is required by Standard 620?

Reply: No.

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Issue Date: April 4, 1986
Publication: Standard 610, Centrifugal Pumps for General Refinery Services,
Edition/Date: Sixth Edition, 1981
Reference: 610-I-01/86

Question 1: Is the maximum unfiltered vibratory displacement of 1.34 mil P.P. correct for a pump with sleeve type bearings operating at 5700 RPM?

Reply: Refer to Figure 3 of paragraph 2.8.2.3. The maximum vibratory displacement would be the intersection of vertical line drawn from 5700 RPM on the abscissa to the 0.4 vibration velocity boundary line. This appears to be closer to 1.5 mils P.P.

Question 2: Which of the following three options is the preferred method to measure the shaft vibration to meet the intent of Standard 610, paragraph 2.8.2.3:

a. Non contact probe mounted thru bearing housing.
b. Non contact probe mounted adjacent to the bearing housing with bracket secured to the pump.
c. Shaft stick (or fish tail).

Reply: The vibration limits as specified by paragraph 2.8.2.3 are for the shaft only as measured from a non-vibratory reference.

Question 3: For routine maintenance and condition monitoring of the pump, what method can be standardized for all Standard 610 pump applications?

Reply: For pumps with antifriction bearings, vibration measurements can be taken at the bearing caps/housing; however, for pumps with sleeve type bearings, vibration measurements at the housings are not a good indiction of the vibratory amplitudes of the shaft. This is due in part by the relative stiffness of antifriction bearings versus sleeve type bearings. Because the method is dependent upon the pump design and configuration, there is no one vibration measurement procedure or method which can give dependable results consistent with the Standard 610 guidelines.

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Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-15/85

Question 1: Is any data or literature available regarding corrosion of crude oil tanks in the Saudi Arabia environment.

Reply: We would refer you to crude oil tank operators in Saudi Arabia and to the section on corrosion in Chapter XIII of the API Guide for Inspection of Refinery Equipment.

Question 2: When should the purchaser specify the corrosion allowance?

Reply: Under "Suggestions for Ordering Tanks" in API Standard 650, corrosion allowance is an option that the purchaser may state in his inquiry or purchase order.

Question 3: Can the purchaser request a corrosion allowance after the contract signature and can he do this without issuing a contract amendment order?

Reply: This is a contractual matter between purchaser and manufacturer.

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E-07-21

1986 INTERPRETATIONS

Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-16/85

Question: Are proof tests for paragraph C.3.4.2 carried out by tank manufacturers in U.S.A. or elsewhere when specified as a requirement by purchaser?

Reply: Yes.

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Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-17/85

Question: Will you incorporate in Standard 650 design rules governing reinforcement and structural stability for large shell openings, up to 100 inches diameter since we require larger openings than presently listed in API 650 in order to install and remove large internals?

Reply: Your request has been referred to the API Subcommittee on Pressure Vessels and Tanks and will be considered during the next revision to the standard.

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1986 INTERPRETATIONS

Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-18/85

Question 1: In using API Standard 650, how does one specify and control construction of the tank bottom in particular the flatness? Is 5.2.2 intended to cover this?

Reply: Paragraph 5.2.2 covers the erection of the bottom of tanks covered by the Scope of API Standard 650.

Question 2: If the peaking section under dimensional tolerances does not include the tank bottom, what does it refer to?

Reply: Paragraph 5.5.3 covers peaking of vertical welds in cylindrical tank shells.

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Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-19/85

Question 1: Are annular bottom plates required if the bottom shell course material is group IV and the bottom material is also Group IV?

Reply: The use of annular bottom plates is related in paragraph 3.5 to the allowable stress used in the design of the bottom shell course.

Question 2: Is it acceptable to post weld heat treat (PWHT) a flush cleanout in the field after tank completion?

Reply: It is the intent of 3.7.7.3 to require PWHT prior to installation. The current paragraph would not prohibit the PWHT after a tank completion. An effective PWHT would require the entire tank to be PWHT which would make this method impractical except on small tanks.

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E-07-21

Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-20/85

Question: For a supported cone roof of an Standard 650 tank is it permissible to weld both ends of the supporting columns to bottom plate and roof member, or is it recommended that the bottom is allowed to remain free and guides provided to permit the columns to float within the guides?

Reply: Paragraph 3.10.4.7 of Standard 650 requires that the column base clip guides be welded to the tank bottom to prevent lateral movement of column bases since the column bases are not usually welded to the tank bottom because of the possibility of bottom settlement. No requirements are stated in Standard 650 regarding welding of the column bases to the tank bottom or to the roof members.

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E-07-21

Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-21/85

Question: What is the range of allowance on the angle of "approximate 90 degree" stated in Figure 3-8 of Standard 650 for the intersection of bottom plate seams with the bottom reinforcing plate?

Reply: No range of tolerance is stated in API Standard 650, but is being given consideration by the subcommittee for possible inclusion in a future edition.

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E-07-21

Issue Date: August 22, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-22/85

Question 1: In Standard 650, is a plate thickness of 6 mm considered the equivalent of 1/4" where paragraph 3.6.1.1 requires a nominal plate thickness of 1/4 inch?

Reply: No.

Question 2: If the 3rd course of a 70.345 ft. diameter by 35.222 ft. high tank is 6 mm rather than 1/4 inch, will the 3rd course be over-stressed per API Standard 650 allowable stresses?

Reply: The American Petroleum Institute does not approve, recommend, or endorse any specific design; nor does it limit the method of design or form of construction that might be used to conform to an API standard. Refer to Standard 650 for your design.

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E-07-21

Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-23/85

Question: Does API Standard 650 specify clearance tolerances between tank bottom plates and foundation?

Reply: No.

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Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-24/85

Question: In a small tank (16 ft. diameter and 16 ft. high) with a self-supporting cone roof, can a frangible joint per paragraph 3.10.2.5(3) be designed?

Reply: In small tanks, the area at the roof-to-shell junction usually exceeds the area calculated by paragraph 3.10.2.5(3) and therefore is not frangible.

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Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-25/85

Question: Is the roof considered frangible if the shell to roof weld is made from the inside?

Reply: No, it is to be made on the top side only as required by 3.10.2.5.

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Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-26/85

Question: Is it permissible on tanks conforming to API Standard 650 to use fabricated/welded angles in lieu of rolled angles as compression rings and top angles per Figure 3-3A?

Reply: Yes, when such fabrication is approved by the purchaser.

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Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-27/85

Question: Do tanks built under Appendixes F and J have to be tested separately according to Appendixes F and J?

Reply: Yes.

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E-07-21

Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-01/86

Question: When a nominal capacity is specified for an API Standard 650 floating roof tank, what actual capacity by volume calculation to the overflow pipe invert would be required?

Reply: The "suggestions for ordering tanks" in API Standard 650 indicates the purchaser should state the nominal capacity with an option to state the diameter and height. A footnote refers to Appendix L, storage tank specification data sheet. The diameter and height should be in conformance with the definitions of D and H in 3.6.3.2 (including Note 1 of 3.6.1.1). The design of floating roof tanks should conform to paragraph C.3 of Appendix C.

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E-07-21

Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-02/86

Question: To fulfill the cross-sectional area requirements of paragraph 3.10.5.2 of Standard 650, must a top angle be provided?

Reply: No, but the participating area requirement of 3.10.5.2 (and 3.10.6.2) shall be satisfied.

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E-07-21

Issue Date: March 10, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-04/86

Question: Do radiographs on 100MM film meet the requirements of Standard 650, paragraph 6.1.2.1?

Reply: Since the conversion of 2 inches to 50MM is within a tolerance of plus or minus 1/10 inch, the implied tolerance at the 2 inch requirement, the radiographs on 100MM film are considered acceptable.

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Issue Date: August 6, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-05/86

Question: Does a lap welded top angle as shown in Figure 3-2 require a continuous weld to the shell both top and bottom?

Reply: Yes.

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Issue Date: December 30, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-06/86

Question 1: In Figure 3-17, Detail E, page 3-39 of the 7th Edition Revision 1 may the 6 inch leg be turned down?

Reply: Yes.

Question 2: Should this be noted in the standard?

Reply: This will be referred to the appropriate subcommittee.

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Issue Date: December 30, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-07/86

Question: If the insert plate around a shell nozzle or manhole (paragraph 6.1.2.1(4)) is provided by a complete shell plate with a thickness equal to that of the calculated insert plate, must the radiographic examination of the vertical and horizontal weld joints of this plate conform to 6.1.2.1(4) or to Figure 6.1 of Standard 650?

Reply: The radiographic examination should conform to Figure 6.1 of API Standard 650.

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Issue Date: December 30, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-08/86

Question: We recommend that the allowable stresses for the second course of large tanks be the same as those of the first course and that the variable design point method be the code rule for calculating the shell minimum thickness of large tanks.

Reply: Thank you for your recommendations. These items have been referred to the appropriate subcommittee.

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Issue Date: December 30, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-09/86

Question: Does 3.1.3.2 of Standard 650 mean one-side or both side welding of 3/16 inch minimum size of fillet welds?

Reply: Paragraph 3.1.3.2 states that minimum size of individual fillet welds. Other paragraphs and figures provide the required details of the welding.

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Issue Date: December 30, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-10/86

Question: Is the only requirement to have a frangible roof a maximum 3/16 inch single fillet weld?

Reply: No. The roof design section of Standard 650 also has requirements.

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Issue Date: December 30, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-11/86

Question: Is a self supported roof of a storage tank considered frangible if it is fabricated in accordance with Detail "A" of Figure "F-1" with:

A. Continuouse single fillet weld (roof to shell)--3/16".
B. Roof slope--2" in 12".
C. Cross-sectional area of roof to shell junction--the value calculated in 3.10.2.5 (Item 3).

Reply: Yes.

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Issue Date: December 30, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-12/86

Question 1: Does 6.1.6 require additional adjacent radiographs if the defect is not within 3 inches of any edge of the film?

Reply: No.

Question 2: Does 6.1.6 require additional adjacent radiographs if the defect is within 3 inches of the edge of the film?

Reply: Yes. Additional radiographs must be taken in the direction of the defect until the limits of unacceptable welding are determined, or the erector may replace all the welding performed by the welder or welding operator on that joint. (See 6.1.6).

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Issue Date: December 30, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980 (ANSI/API 650-1980)
Reference: 650-I-15/86

Question: In Standard 650, is it necessary to consider group numbers included in Table QW 422 of ASME Section IX in qualifying welding procedures requiring notch toughness?

Reply: No.

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Issue Date: September 4, 1986
Publication: Standard 650, Welded Steel Tanks for Oil Storage
Edition/Date: Seventh Edition, 1980, Revision 1, February 1984

Question 1: Is gas blanketing a requirement so that the outer tank can be sealed? If so, why? (Appendix H, H.6.2.2 items 2 and 3)

Reply: Gas blanketing is not a requirement under Standard 650, Appendix H. However, it could be deemed appropriate by particular companies under certain circumstances such as a trade off for tank spacing requirements. When gas blanketing is utilized on an internal floating roof tank, it is usually for the purpose of achieving complete isolation of the tank contents from the atmosphere. In effect, the resulting blanket is a type of seal.

In the petroleum industry, a gas blanket would prevent product contamination from oxygen or condensation from moisture in the atmosphere. If this is of real concern, it would normally be more cost effective to provide a conventional cone roof tank with gas blanketing.

In some isolated cases, there may be some incentive to use an internal floater when the authority having jurisdiciton requires greater tank spacing or special fire protection for a gas blanketed cone roof tank.

Question 2: If gas blanketing is not done, what special safety precautions are needed to ensure a safe installation?

Reply: In the absence of gas blanketing, special safety precautions needed to ensure a safe installation include:

1. Periodic checks of the atmosphere above the floating roof deck to ensure that the seals are in good condition.
2. Safe operating practices which avoid routinely lowering the liquid level so the tank deck rests on its support thus creating a vapor space under the deck. The latter practice will in the case of flammable liquids, result in a flammable atmosphere above the deck during the initial filling cycle. In this situation the release of vapors at the open vents could be hazardous if lightning occurred simultaneously with the filling operation.

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Question 3: Do you know of any petroleum companies who have sealed the outer tank with a PV vent but do not perform gas blanketing?

Reply: The only cases we are aware of where the outer tank was sealed with a PV vent without performing gas blanketing consist of instances where an existing cone roof tank was converted to an internal floating roof tank. We would like to point out that most of our members do not subscribe to this practice and it does not conform wih Standard 650.

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Issue Date: May 23, 1986
Publication: Standard 674, Positive Displacement Pumps--Reciprocating
Edition/Date: First Edition, 1980
Reference: 674-I-01/84

Question 1: Is Standard 674 applicable to pumps with horizontal plungers? Is it also applicable to pumps with vertical plungers?

Reply: The user may specify, or the vendor may offer, either horizontal or vertical cylinders on pumps meeting the requirements of Standard 674. Not all vendors make both types.

Question 2: In order to calculate the maximum speed in a continuous operation, we would like to know if the API Standard 674 Tables 1 and 2 and viscosity correction factor of paragraph 2.3.3 are applicable or not on pumps with vertical plungers?

Reply: Tables 1 and 2 and paragraphs 2.3.3 of Standard 614 apply to pumps with either horizontal or vertical cylinders.

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Date: April 24, 1986
Publication: Publication 941, Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants
Edition/Date: Third Edition, 1983
Reference: 941-I-01/86

Question: In view of the questions sporadically raised by our staff members and by customers and with reference to your publication regarding the use of 0.5% Mo steel for hydrogen service, we would ask you to inform us whether new findings have been obtained in recent years, by virtue of which the curve is modified or the 0.5% Mo steel figure deleted from Standard 941. In case of an affirmative answer, please state whether Publication 941 will be revised shortly.

Reply: Since the issuing of the May 1983 Third Edition of API Publication 941, the API has received additional information where 0.5% Mo alloy steel has experienced hydrogen attack below the present 0.5% Mo line. This information is currently being reviewed by the appropriate technical committee. It is expected that a revision to the third edition incorporating all new data and reflecting any modifications to the 0.5% Mo line will be issued.

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Issue Date: August 26, 1986
Publication: Publication 2202, Dismantling and Disposing of Steel from Tanks Which Have Contained Leaded Gasoline
Edition/Date: Second Edition, 1982

Question: Does 4.2 apply to all storage tanks that have contained or may be contained leaded gasoline?

Reply: Publication 2202 states, "If the steel is to be sold for scrap and possible smelting in the manufacture of new steel, all interior surfaces of any plates that have been in contact with sludge should be abrasive blasted or wire brushed to bare metal."

It is the opinion of the subcommittee members having jurisdiction over the publication that the procedures in Publication 2202 were prepared specifically for large aboveground storage tanks and not for smaller underground tanks.

Contact with the mnembers who reviewed this publication in 1982 and 1976 indicates that at no time were dismantling and disposal procedures for smaller undergroud tanks discussed. A review of our minutes over the past five years verifies this opinion. The subcommittee believes indications of the publication's intent are found throughout the text. Examples are:

1. The word "dismantling" in the title and elsewhere. Smaller tanks usually do not require dismantling.
2. Reference to "dismantlement" from inside the tanks.
3. Reference to the "bottom ring of the shell and plates from the tank bottom," "roof plates," and "roof supporting columns," which are not found in smaller underground tanks.

Scale and sludge inside large petroleum storage tanks can contain lead contamination will after the tank is removed from service. However, our member companies experiences with small underground retail tanks having 8000 gallon capacities indicate there have been no problems with scaling or sludge accumulation. The following factors support these experiences:

- o The accumulation of sludge in small underground tanks tends to be minimized by frequent filling and emptying and by the infrequent presence of water bottoms.
- o The gasoline stored in retail outlet tanks is always run through strainers at the terminal, frequently filtered, and always delivered in clean tank trucks. Consequently, there is minimal chance for sludge accumulation.
- o Retail underground tanks remain at more uniform temperature, and thus breathe much less than aboveground tanks subject to

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diurnal heating and cooling. This reduces moisture intake and thereby minimizes rust and wall scale.

- Current retail operating practices require that each underground tank be checked for the presence of water before and after each delivery. Most companies require removal of water if the level exceeds approximately 1/2 inch.

In summary, we believe that the level of lead contamination that may occur to steel in underground storage tanks is sufficiently low as to not require the same level of handling as presently required in Publication 2202, paragraph 4.2 for bulk petroleum storage tanks.

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Issue Date: December 4, 1986
Publication: Bulletin 2514; Publication 2514A, Atmospheric Hydrocarbon Emissions from Marine Vessel Transfer Operations
Edition/Date: November 1959; September 1981

Question: Does the data presented in Bulletin 2514 and its successor, Publication 2514A allow the correct calculation of in-transit atmospheric emissions?

Reply: API Bulletin 2514, November 1959

Bulletin 2514 (which has been withdrawn and replaced by API Publication 2514A) did include a discussion of in-transit losses. As described on page 14 of that bulletin: "A limited amount of data received on marine transit loss are included for orientation purposes."

We believe tht this data was not intended to substantiate a loss rate, but to simply present some preliminary data. Additionally at the time this data was collected, crude oil tankers were not required to use inert gas blanketing during the 1970s. These systems require that a relatively tight seal be maintained on cargo compartments so that the inert gas blanket is maintained. Therefore, the data published in API 2514 (1959) should not be used in estimating in-transit breathing losses from crude oil tankers, especially tankers with inert gas systems.

API Publication 2514 A, September 1981

The most up-to-date discussion by API of in-transit breathing losses on crude oil tankers is presented in Publication 2514A, Second Edition, September 1981. The discussion of the breathing losses is in Appendix E:

"Evaporation cargo loss occurs whenever a liquid carge evaporates, regardless of whether or not the vapor is displaced from the compartment and emitted into the atmosphere. Evaporation and, thus, cargo loss, occurs whenever a volatile liquid comes into contact with air (or an inert gas) that is not saturated with hydrocarbon vapor. This occurs primarily during loading and discharge operations for non-inerted vessels, operating at essentially atmospheric pressure. Other operations, such as manual tank gaging and vessel transit, results in relatively neglible losses. This judgment is based on estimating typical breathing loses, assuming operating pressure/vacuum valves, and calculating losses from tank clingage, based on empirical clingage factors. Therefore, total evaporative cargo loss for vessels operated at

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atmospheric pressure can be estimated by summing the losses that occur during loading and discharge."

This paragraph makes several points important to a discussion of in-transit emissions to the atmosphere. First, the loss cacu-lations in the API publications are intended to approximate the loss of liquid cargo to the tanker's blanketing gas. Second, the loss of liquid cargo is not necessarily displaced from the cargo compartment and emitted into the atmosphere. Third, in-transit cargo losses are relatively negligible.

We realize that environmental agencies have used the data in Publication 2514A to estimate atmospheric emissions during tanker loading. This method has validity, since during tanker loading a known volume of the tanker blanketing gas (which contains hydro-carbon from cargo loss) is displaced to the atmosphere by the volume of crude oil loaded into the tanker.

On the other hand, using the API data to calculate atmospheric emissions during transit lacks validity. First, the API does not present any data for use in estimating in-transit cargo loss (which is loss to the blanketing gas) from crude oil tankers with inert gas systems. Second, and we feel more importantly, the API has never presented any data which enables an estimate of the blanketing gas that is discharged to the atmosphere while a tanker is in transit.

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